Karol Pietruszka

Geospatial Data Analyst & Forest Engineer

Kendlerstraße 6/14, 1140 Vienna, Austria | +43 681 81130013 | karolpietruszka88@gmail.com

linkedin.com/in/karol-pietruszka-8b4645236 | karolpiet.github.io/Portfolio | github.com/karolpiet

SUMMARY

Geospatial Data Analyst and Forest Engineer with postgraduate education in Geographic Information Systems (UNIGIS). Over 10 years of professional experience in forestry, spatial data analysis, and remote sensing, with a strong focus on LiDAR point cloud processing, satellite imagery analysis, and GIS-based forest management.

Experienced in developing and automating geospatial workflows, processing large-scale environmental datasets, and delivering practical analytical solutions for forestry and land management applications.

SKILLS

- **GIS & Remote Sensing:** ArcGIS Pro (Spatial Analyst, Network Analyst), ArcMap, QGIS, ERDAS Imagine, Google Earth Engine, Sentinel-2 satellite imagery, NDVI / NBR / dNBR burn severity analysis, land cover classification and change detection, map composition.
- **LiDAR:** point cloud filtering and classification, Digital Terrain and Surface Models (DTM/DSM), terrestrial / mobile point clouds (.las/.laz), tree detection, segmentation and DBH measurement.
- **Programming & Data:** Python (ArcPy, GeoPandas, Pandas, NumPy, Rasterio, Shapely, GDAL, laspy, scikit-learn, geemap), SQL, Jupyter Notebook, JavaScript (Leaflet, three.js), HTML / CSS, Git & GitHub.
- **Core Competencies:** Spatial Data Quality Control, Automated Cartography, Independent Project Management, Cross-functional Team Collaboration, High Stress Resistance under Tight Deadlines.
- **Office Suite:** MS Office.
- **Languages:** Polish (Mother tongue), English (B2 – Professional Working Proficiency), German (A2 – Elementary Proficiency).
- **Driver's license:** Cat. A, B, C.

SELECTED PROJECTS

Details and interactive maps: karolpiet.github.io/Portfolio

- **LidarTreeSegmenter (in development):** Python tool for tree detection, segmentation and forest inventory (DBH, height, volume) from terrestrial / mobile LiDAR point clouds; outputs Excel report, GeoPackage with QGIS project and interactive 3D view.
- **Forest fire severity – Bohemian Switzerland National Park:** Sentinel-2 and Google Earth Engine dNBR burn severity mapping of the 2022 wildfire, restricted to forest pixels with ESA WorldCover.
- **Land cover change – Bory Tucholskie:** Supervised classification and NDVI change detection (2015–2019) after the 2017 windstorm in ERDAS Imagine; web map in Leaflet.
- **Coastal flood risk – Poland:** Population at risk from a 5 m sea level rise using EU-DEM and GHSL data and ArcGIS Pro Spatial Analyst.
- **Bicycle accessibility – Wrocław:** Network analysis (Service Area) of 5, 10 and 15-minute cycling ranges to railway stations using OpenStreetMap data.

WORK EXPERIENCE

GIS Technologist & Forest Assessor 07.2024 – now

Umweltdata GmbH (Austria)

- Process and validate large-scale geospatial datasets to generate digital forest maps, management models, and district plans.
- Utilize remote sensing data, high-resolution orthophotos, and satellite imagery to design custom GIS solutions.
- Manage and structure complex forestry field data, ensuring high database consistency and quality control.

Landscape & Garden Design Technician 03.2024 – 07.2024

Vicena (Vienna, Austria)

- Executed professional urban garden layouts and green area maintenance according to spatial planning guidelines.

Geospatial Data Analyst & Forest Assessor 10.2015 – 04.2023

The Bureau for Forest Management and Geodesy (Cracow, Poland)

- Automated the filtering, classification, and analysis of LiDAR point clouds to generate Digital Terrain and Surface Models.
- Modeled and designed target forest road networks using advanced vector and spatial analysis in ArcMap and QGIS.
- Performed precise editing, data ingestion, and quality management of geodata for comprehensive national-level forest inventories.
- Delivered cartographic compositions and key project data for high-stakes initiatives, including the National Forest Inventory and specialized research for the Forest Research Institute (Biostrateg, Monitoring Beskidy).

Forest Management Specialist / Assessor 08.2013 – 09.2015

Taxus SI – Department for Forest Management (Poland)

- Worked on site on the projects creating full and simplified forest inventories and collecting spatial data.

Green Space Development Specialist 11.2012 – 08.2013

SMREK (Poland)

- Developed, planned, and arranged urban and rural green areas in line with regional spatial requirements.

Graduate Forestry Intern 10.2011 – 10.2012

State Forests, Daleszyce Forest District (Poland)

- Gained practical hands-on experience in daily forest administration, silviculture, timber harvesting, and logistics management.

EDUCATION

Postgraduate Studies: Geomatics and Artificial Intelligence in Forestry 2026 – now

Forest Research Institute (Poland)

Postgraduate Diploma: UNIGIS Geographic Information Systems (GIS) 2022 – 2023

Jagiellonian University, Institute of Geography and Spatial Management (Cracow, Poland)

Engineer in Forestry (Focus: Forest Management) 2010 – 2016

University of Agriculture (Cracow, Poland)

Engineering thesis: Monetary value of timber resources in "Uroczysko Kowalow Las" in Beskid Niski

Certified Forest Technician 2004 – 2008

Forest Technical School in Bałtów (Poland)

COURSES

- **ESRI Academy:** Certificates confirming completion of various courses and advanced workflows in the ArcGIS ecosystem.
- **Udemy:** Python from basics to expert – focus on data analysis, programming syntax, and automation scripting.

INTERESTS

- **Technology and Innovation:** Particularly interested in the development of GIS tools and their applications in various engineering fields.
- **Nature:** Environmental observation and forestry, spending time outdoors to maintain a healthy work-life balance.
- **Automotive:** Passionate about motorcycle touring, enjoying both short and long-distance travel.